Homework #1 solution (total 100 pts.)

1 (25 pts.) Determine whether the system described by the differential equation

$$\frac{d^2x}{dt^2} + (6 + 3\,sin\,t)x = 0$$

is linear or nonlinear by checking the <u>homogeneity</u> and the <u>additivity</u> properties.

(Units: x [m], t [s])

Homogenity:

$D(\alpha x)$

$= \frac{d^2(\alpha x)}{dt^2} + (6 + 3\,sin\,t) \cdot (\alpha x)$

$= \alpha \cdot \frac{d^2(x)}{dt^2} + \alpha \cdot (6 + 3\,sin\,t) \cdot (x)$

$= \alpha \left(\frac{d^2(x)}{dt^2} + (6 + 3\,sin\,t)x \right)$

$= \alpha D(x)$

∴ Homogenity is satisfied.

Additivity:

$D(x_1 + x_2) = \frac{d^2(x_1 + x_2)}{dt^2} + (6 + 3\,sin\,t) \cdot (x_1 + x_2)$

$D(x_1) + D(x_2) = \frac{d^2(x_1)}{dt^2} + (6 + 3\,sin\,t) \cdot (x_1) + \frac{d^2(x_2)}{dt^2} + (6 + 3\,sin\,t) \cdot (x_2)$

$D(x_1 + x_2) = D(x_1) + D(x_2)$

∴ Additivity is satisfied.

Satisfying homogeneity & additivity → linear system

2 (25 pts.) A system is described by the differential equation

$$\frac{d^2x}{dt^2} + 0.6\frac{dx}{dt} + 9x = 10\,sin\,\omega t\,.$$

Determine the dynamic stiffness, the frequency response, and the system response.

Then plot the magnitude and the phase angle of the frequency response.

(Units: x [m], t [s], $10sin\,\omega t\,[N]$)

Dynamic stiffness : $Z(j\omega) = (j\omega)^2 + 0.6j\omega + 9$

Frequency response : $G(j\omega) = \frac{1}{(9-\omega^2)+0.6j\omega}$

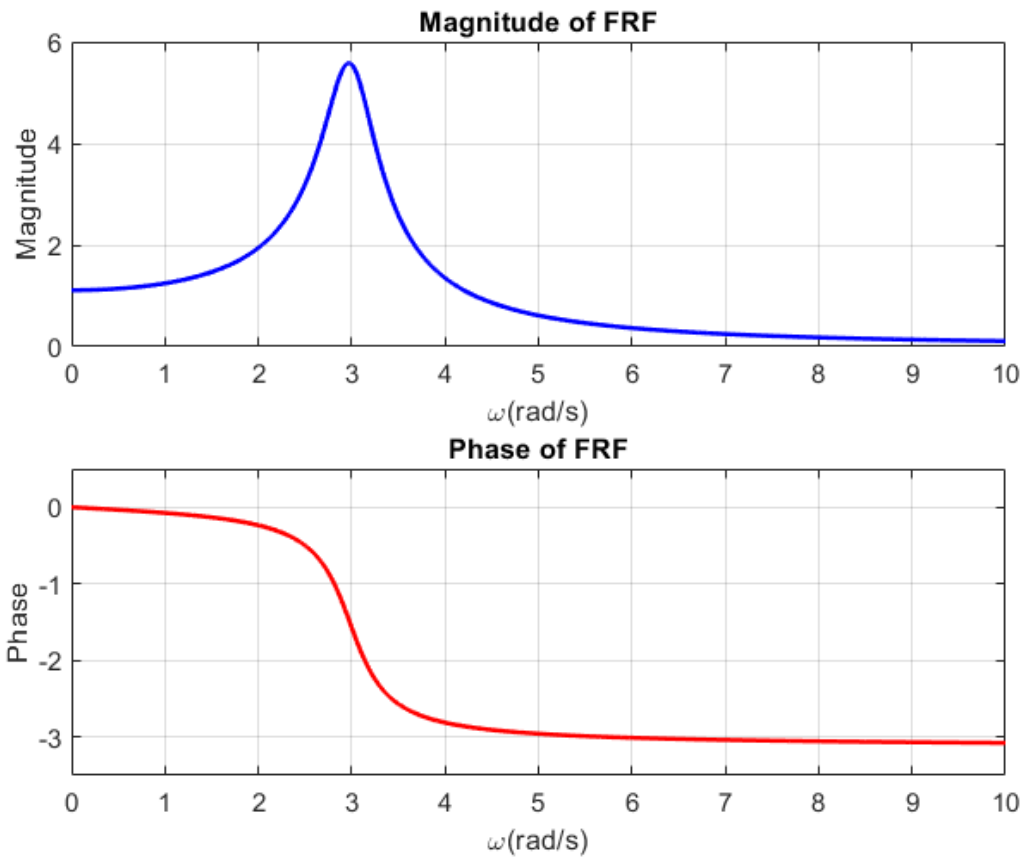
System response :

$x(t) = G(j\omega)F_0e^{j\omega t}$

$= Re\left[\frac{10e^{j\omega t}}{(9-\omega^2)+0.6j\omega}\right]$

$= \frac{10}{\sqrt{(9-\omega^2)^2+(0.6\omega)^2}}\sin(\omega t - \phi)$

$where\ \phi = tan^{-1}\left(\frac{0.6\omega}{9-\omega^2}\right)$



3 (25 pts.) Derive the <u>impulse response</u> for the system described by the differential equation

$$\frac{d^3x}{dt^3} + 2\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = \delta(t)\,.$$

And derive the proper <u>state equation</u>($\dot{\mathrm{x}}(\mathrm{t}) = \mathbf{A}\mathrm{x}(\mathrm{t}) + \mathbf{b}f(\mathrm{t})$) of the same system.

(Units: x [m], t [s], δ $[N]$)

System response:

$\frac{d^3x}{dt^3} + 2\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = \delta(t)$

→ Laplace transform

$(s^3 + 2s^2 + 4s + 8)X(s) = 1$

$X(s)$

$= \frac{1}{s^3+2s^2+4s+8}$

$= \frac{1}{(s+2)(s^2+4)} = \frac{A}{(s+2)} + \frac{Bs+C}{(s^2+4)}$; $A = \frac{1}{8}$, $B = -\frac{1}{8}$, $C = \frac{1}{4}$

→ Inverse Laplace transform

$x(t) = \frac{1}{8}e^{-2t} - \frac{1}{8}\cos 2t + \frac{1}{8}\sin 2t \quad (t > 0)$

State equation:

$\frac{d^3x}{dt^3} + 2\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = \delta(t)$ → $\dddot{x}(t) = -2\ddot{x}(t) - 4\dot{x}(t) - 8x(t) + \delta(t)$

$$\dot{\mathbf{x}}(\mathrm{t}) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -4 & -2 \end{bmatrix} \mathbf{x}(\mathrm{t}) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \delta(t)\,; \quad \text{where } \mathbf{x}(\mathrm{t}) = \begin{bmatrix} x(\mathrm{t}) \\ \dot{x}(\mathrm{t}) \\ \ddot{x}(\mathrm{t}) \end{bmatrix}$$

4 (25 pts.) Derive the <u>step response</u> for the system described by the differential equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 9x = u(t)$$

by means of the approach based on the transition matrix.

(Units: x [m], t [s], u [N])

State equation:

$$\dot{\mathbf{x}}(\mathrm{t}) = \begin{bmatrix} 0 & 1 \\ -9 & -4 \end{bmatrix} \mathbf{x}(\mathrm{t}) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)\,; \quad \text{where } \mathbf{x}(\mathrm{t}) = \begin{bmatrix} x(\mathrm{t}) \\ \dot{x}(\mathrm{t}) \end{bmatrix}$$

$$e^{\boldsymbol{A}t} = \begin{bmatrix} e^{-2t}\cos(\sqrt{5}t) + \frac{2\sqrt{5}}{5}\sin(\sqrt{5}t) & \frac{\sqrt{5}}{5}e^{-2t}\sin(\sqrt{5}t) \\ -\frac{9\sqrt{5}}{5}e^{-2t}\sin(\sqrt{5}t) & e^{-2t}\cos(\sqrt{5}t) - \frac{2\sqrt{5}}{5}\sin(\sqrt{5}t) \end{bmatrix}$$

Letting $\mathbf{x}(0)=\mathbf{0}$,

$$\mathbf{x}(\mathrm{t}) = \int_0^{\mathrm{t}} e^{\mathbf{A}(\mathrm{t}-\tau)}\mathbf{b}\mathrm{u}(\tau)\,\mathrm{d}\tau$$

$$= \int_0^{\mathrm{t}} e^{\mathbf{A}(\mathrm{t}-\tau)} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mathrm{u}(\tau)\,\mathrm{d}\tau$$

$$= \int_0^{\mathrm{t}} \begin{bmatrix} \frac{\sqrt{5}}{5}e^{-2(t-\tau)}\sin\left(\sqrt{5}(t-\tau)\right) \\ e^{-2(t-\tau)}\cos\left(\sqrt{5}(t-\tau)\right) - \frac{2\sqrt{5}}{5}\sin\left(\sqrt{5}(t-\tau)\right) \end{bmatrix} \mathrm{u}(\tau)\,\mathrm{d}\tau\ ; \quad \text{where } \mathbf{x}(\mathrm{t}) = \begin{bmatrix} x(\mathrm{t}) \\ \dot{x}(\mathrm{t}) \end{bmatrix}$$

$$\rightarrow\ x(t) = \int_0^{t} \frac{\sqrt{5}}{5}e^{-2(t-\tau)}\sin\left(\sqrt{5}(t-\tau)\right)\mathrm{u}(\tau)\,\mathrm{d}\tau$$

$$\therefore\ x(t) = \frac{1}{9} - \frac{1}{9}e^{-2t}\,cos(\sqrt{5}t) - \frac{2\sqrt{5}}{45}e^{-2t}\,sin(\sqrt{5}t)\ ;\ \ (t > 0)$$

cf) Apply the Laplace transform with $\mathcal{L}[\mathbf{x}(\mathrm{t})] = \boldsymbol{X}(s)$.

$$\boldsymbol{sX}(s) = \boldsymbol{AX}(s) + \mathcal{L}[\mathbf{u}(\mathrm{t})]$$

$$\rightarrow (s\boldsymbol{I} - \boldsymbol{A})\boldsymbol{X}(s) = \mathcal{L}[u(t)] = \begin{bmatrix} 0 \\ 1 \end{bmatrix}\frac{1}{s} \qquad \left(\because\ Step\ response:\ \mathcal{L}[u(t)] = \frac{1}{s}\right)$$

$$\boldsymbol{X}(s) = (s\boldsymbol{I} - \boldsymbol{A})^{-1}\begin{bmatrix} 0 \\ 1 \end{bmatrix}\frac{1}{s} = \begin{bmatrix} s & -1 \\ 9 & s+4 \end{bmatrix}^{-1}\begin{bmatrix} 0 \\ 1 \end{bmatrix}\frac{1}{s} = \frac{1}{s(s+4)+9}\begin{bmatrix} s+4 & 1 \\ -9 & s \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}\frac{1}{s} = \frac{1}{s(s^2+4s+9)}\begin{bmatrix} 1 \\ s \end{bmatrix}$$

$$\rightarrow \boldsymbol{X}_1(s) = \frac{1}{s(s^2+4s+9)} \rightarrow \mathcal{L}^{-1}[\boldsymbol{X}_1(s)] = \mathcal{L}^{-1}\left[\frac{1}{9}\left(\frac{1}{s} - \frac{s+2}{(s+2)^2+(\sqrt{5})^2} - \frac{2\sqrt{5}}{5}\frac{\sqrt{5}}{(s+2)^2+(\sqrt{5})^2}\right)\right]$$

$$\rightarrow x(t) = \frac{1}{9} - \frac{1}{9}e^{-2t}\,cos(\sqrt{5}t) - \frac{2\sqrt{5}}{45}e^{-2t}\,sin(\sqrt{5}t)\ ;\ \ (t > 0)$$